

From-Scratch Replication Report:

Chiral and Topological Orbital Magnetism of Spin Textures

Lux, Freimuth, Blügel, Mokrousov, arXiv:1706.06068 (2017)

Hermes Agent — REPLICATE-PROJECT / TEXTURES-100

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Abstract

Headline claim (paper Eq. 12): at *zero* spin–orbit coupling the topological orbital magnetization is

$$M_{\text{tom}} = \frac{1}{4} \chi_{\text{LP}}^{\uparrow+\downarrow} B_{\text{eff}}^z \text{sgn}(\Delta_{xc}) (1 - 3\mu^2/\Delta_{xc}^2), \text{ i.e. TOM is linear in the scalar spin chirality}$$

$$B_{\text{eff}}^z = \frac{\hbar}{2e} \hat{n} \cdot (\partial_x \hat{n} \times \partial_y \hat{n}).$$

This work: 2D square-lattice *s*–*d* model (*no SOC*) with a Néel skyrmion; itinerant $L_z = \frac{1}{2}(Xv_y - Yv_x)$. We CONFIRM the structural headline: $M_{\text{tom}} \propto$ scalar spin chirality (linear fit $R^2 = 0.964$, M_{tom}/χ constant to 13% across textures). The μ -dependence sign change is reproduced qualitatively (PARTIAL); the absolute $\frac{1}{4}\chi_{\text{LP}}$ continuum prefactor is a normalization gap.

Verdict: PARTIAL. Coverage 6/10 Agreement 7/10.

1 The paper’s claim

Lux *et al.* use a semiclassical Green’s-function / Wigner gradient expansion to derive corrections to the out-of-plane orbital magnetization of a locally ferromagnetic 2D system, ordered in powers of the gradients of the magnetization $\hat{n}(x, y)$: a *chiral* orbital magnetization (COM, first order, Eq. 2) and a *topological* orbital magnetization (TOM, second order, Eq. 3). Evaluated in the 2D Rashba model (Eq. 7), the central result is the **zero-SOC** TOM:

$$M_{\text{tom}} = \frac{1}{4} \chi_{\text{LP}}^{\uparrow+\downarrow} B_{\text{eff}}^z \text{sgn}(\Delta_{xc}) \left(1 - 3 \frac{\mu^2}{\Delta_{xc}^2}\right), \quad |\mu| < |\Delta_{xc}|, \quad (1)$$

with the emergent field / scalar spin chirality $B_{\text{eff}}^z = \frac{\hbar}{2e} \hat{n} \cdot (\partial_x \hat{n} \times \partial_y \hat{n})$ (Eq. 1) and the diamagnetic Landau–Peierls susceptibility $\chi_{\text{LP}}^{\uparrow+\downarrow} = -e^2/(12\pi m_e)$. The comparison anchors are: **(H1)** TOM is *linear in the scalar spin chirality* with a 1/4 coefficient (vs the COM’s 1/2, Eq. 11); **(H2)** the μ -factor $(1 - 3\mu^2/\Delta_{xc}^2)$ produces a sign change at $|\mu| = |\Delta_{xc}|/\sqrt{3}$ and a vanishing beyond $|\mu| = |\Delta_{xc}|$.

2 Method built (from scratch)

No author code exists. We rebuild the physics as a real-space lattice problem rather than the continuum Green’s-function expansion, so that the same physical model can be diagonalized directly. Model (*NO SOC term anywhere*):

$$H = -t \sum_{\langle ij \rangle} c_i^\dagger c_j + m \sum_i c_i^\dagger (\hat{n}_i \cdot \vec{\sigma}) c_i, \quad (2)$$

with $\hat{n}_i$ a Néel-skyrmion texture. The itinerant (atomic-center) orbital operator, following the shared göbel2024 kernel, is

$$L_z = \frac{1}{2}(Xv_y - Yv_x), \quad v_a = i[H, R_a], \quad (3)$$

center-gauged ($R \rightarrow R - R_{\text{center}}$, pitfall 8). The itinerant topological orbital magnetization is the accumulated ground-state moment $M_{\text{orb}} = \text{Tr}[PL_zP]/N_{\text{cell}}$ with P the projector onto occupied states at μ , and the texture contribution is isolated by subtracting the uniform-FM reference at matched filling, $M_{\text{tom}} = M_{\text{tex}} - M_{\text{FM}}$.

The scalar spin chirality is measured two ways: the Berg–Lüscher lattice solid angle (topologically quantized $\rightarrow 4\pi N_{\text{sk}}$, used as a sanity check), and the *continuum* finite-difference density $\chi_c = \sum_x \hat{n} \cdot (\partial_x \hat{n} \times \partial_y \hat{n})$, which tracks the local emergent-field density B_{eff}^z and varies continuously as the texture is canted — this is the quantity Eq. 12 couples to.

Kernel credit. The lattice s - d Hamiltonian builder (`build_H`, `build_FM`), the skyrmion texture, and the itinerant $L_z = \frac{1}{2}(Xv_y - Yv_x)$ operator are reused from Ollie’s shared kernel `gobel2024.sd.skyrmion.kubo.Lz.kernel.py` (göbel2024, arXiv:2410.00820). Our from-scratch additions: an arbitrary-texture Hamiltonian builder, the collinear $\rightarrow$ skyrmion canting sweep, the continuum scalar-spin-chirality estimator, the center-gauged ground-state orbital moment, and the H1/H2 comparison harness.

3 Results

3.1 H1 — TOM is linear in the scalar spin chirality (CONFIRMED)

Canting a fixed Néel skyrmion ($\lambda = 3$, $L = 24$, $m = 3t$, filling 0.06, $\alpha_R = 0$) from collinear ($\hat{z}$) to the full texture dials the scalar spin chirality χ_c continuously. The FM-subtracted itinerant TOM tracks it linearly:

η (canting)	χ_c (scalar chirality)	M_{tom}	M_{tom}/χ_c
0.15	−0.0007	-9.2×10^{-5}	— (noise floor)
0.35	−0.156	-7.3×10^{-4}	—
0.50	−2.790	-2.08×10^{-3}	7.5×10^{-4}
0.65	−7.930	-4.29×10^{-3}	5.4×10^{-4}
0.80	−9.847	-6.21×10^{-3}	6.3×10^{-4}
1.00	−10.764	-8.00×10^{-3}	7.4×10^{-4}

Linear fit $M_{\text{tom}} = s\chi_c + b$: $s = 6.3 \times 10^{-4}$, intercept $b = -2.6 \times 10^{-4}$ (small vs the data span), $R^2 = 0.964$. Over appreciable-chirality points M_{tom}/χ_c is constant to **13%** (relative spread 0.129). This directly reproduces the paper’s structural headline: *at zero SOC the topological orbital magnetization is proportional to the scalar spin chirality* (Eq. 12, $M_{\text{tom}} \propto B_{\text{eff}}^z$). The response is *purely texture-induced* — there is no SOC term in H , and $M_{\text{FM}} = 0$ on the collinear reference.

3.2 H2 — μ -dependence (PARTIAL)

Sweeping μ across the lower band at fixed skyrmion, $M_{\text{tom}}(\mu)$ is a *sign-changing, non-monotonic* function (5 sign changes over the sampled range), qualitatively consistent with the paper’s statement that TOM changes sign and is strongly μ -dependent. However the *clean single parabola* ($1 - 3\mu^2/\Delta_{xc}^2$) of Eq. 12 is a near-band-edge *continuum* result; the full-band lattice sweep is dominated by van Hove / finite-size band structure, so we resolve only the qualitative sign change, not the clean parabolic shape.

4 Comparison to the 1/4 coefficient

The paper’s coefficient is $\frac{1}{4}\chi_{\text{LP}}^{\uparrow+\downarrow}$ with $\chi_{\text{LP}} = -e^2/(12\pi m_e)$ the *continuum* Landau–Peierls susceptibility of a parabolic band. Our extracted slope $M_{\text{tom}}/\chi_c \approx 6.7 \times 10^{-4}$ is in *lattice/tight-binding* units (itinerant $\frac{1}{2}(r \times v)$ operator, $t = 1$), which do not map onto the continuum χ_{LP} without an operator-normalization bridge. We therefore **do not** claim numerical agreement on the absolute 1/4 prefactor; we confirm the two *structural* statements that the coefficient encodes: (i) *linearity* in the scalar spin chirality, and (ii) the sign of the response. This is the same modern-theory-of-orbital-magnetization normalization limit documented for the sibling göbel2024/2025 skyrmion-OHE runs (pitfalls 8/11): the qualitative/structural physics reproduces, the strict continuum prefactor requires the Berry-phase orbital-magnetization operator.

5 Gaps, conventions and scope

- **Absolute $\frac{1}{4}\chi_{\text{LP}}$ coefficient:** continuum-vs-lattice normalization gap (above); structural linearity confirmed, absolute prefactor not matched.
- **FM subtraction** is a difference of finite orbital moments (pitfall 8); here $M_{\text{FM}} = 0$ for the collinear reference, so the H1 residual is clean.
- **COM / SOC branch** (Eqs. 8, 11) and the full Fig. 2 (α_R, Δ_{xc}) phase diagram were not rebuilt — out of scope for the *zero-SOC* headline.
- **Method substitution:** continuum semiclassical Green-function gradient expansion $\rightarrow$ direct lattice diagonalization of the same physical model.

Verdict

PARTIAL — the zero-SOC structural headline ($\text{TOM} \propto$ scalar spin chirality, correct sign, linear response) is reproduced ($R^2 = 0.964$); the clean μ -parabola and the absolute $1/4\chi_{\text{LP}}$ coefficient are not. Coverage 6/10, Agreement 7/10.

Reproduce. `/home/stevens/comfyui-env/bin/python work/lux2017_tom.py` (runtime ≈ 18 s; writes `work/lux2017_result.json`).